6.Query Expansion and Relevance Feedback

Query expansion is used to improve search results by following local methods:

1) Relevance Feedback (Roccio Algorithm)
2) Pseudo Relevance Feedback
   a) Association cluster
   b) Metric cluster
   c) Scalar cluster

The following formula is used to implement Roccio algorithm:

- Used in practice:

$$\vec{q}_m = \alpha\vec{q}_0 + \beta\frac{1}{|D_r|}\sum_{\vec{d}_j \in D_r}\vec{d}_j - \gamma\frac{1}{|D_{nr}|}\sum_{\vec{d}_j \in D_{nr}}\vec{d}_j$$

- D_r = set of known relevant doc vectors
- D_{nr} = set of known irrelevant doc vectors
  - Different from C_r and C_{nr} !
- q_m = modified query vector; q_0 = original query vector; α,β,γ: weights (hand-chosen or set empirically)
- New query moves toward relevant documents and away from irrelevant documents

According to this image, 20 queries were fed as q_0 as input and the algorithm was tested for alpha =1.0, beta = 0.9 and gamma =0.1 values , so that a set of weights could be found that can separate relevant documents from the irrelevant documents.

The 20 queries were selected based on number of query parameters, spelling mistakes, regional name of countries and queries containing words that are states or cities (outside of country domain).

1) 4 queries with one keyword for the data that was crawled
   a) australia
   b) china
   c) ethiopia
   d) pakistan
2) 4 queries with two keywords for the data that was crawled
   a) bangladesh area
   b) pakistan terrorism
   c) china population
   d) india pakistan
3) 4 queries with spelling mistakes in the query words for the data that was crawled
   a) australai
   b) paksitan
   c) chna
   d) germny
4) 4 queries with regional name for countries
   a) hindustan
   b) britaan
   c) franche

d) gana

5) 4 queries with the words that are outside of country domain
a) texas
b) europe
c) mumbai
d) doha

examples of the web pages that are found relevant:

Query	**Relevant web pages**
australia	https://en.wikipedia.org/wiki/Australia
	https://www.britannica.com/place/Australia
	https://en.wikipedia.org/wiki/Demographics_of_Australia
	https://en.wikipedia.org/wiki/Indigenous_Australian_music
china population	https://worldpopulationreview.com/countries/china-population/
	https://en.wikipedia.org/wiki/Chinese_people
	https://en.wikipedia.org/wiki/One-child_policy
	https://www.statista.com/topics/753/china/
india pakistan	https://en.wikipedia.org/wiki/Pakistan_Movement
	https://en.wikipedia.org/wiki/Kashmir_conflict
	https://worldview.stratfor.com/topic/india-pakistan-rivalry

examples of the web pages that are found irrelevant:

Query	**Irrelevant web pages**
australia	https://en.wikipedia.org/wiki/Cocos_Islands
	https://en.wikipedia.org/wiki/Demographics_of_Cocos_(Keeling)_Islands
	https://en.wikipedia.org/wiki/Applied_linguistics
china population	https://en.wikipedia.org/wiki/Taiwan,_China
	http://www.spacetoday.org/China/ChinaGlance.html
	https://www.thoughtco.com/peoples-republic-of-china-facts-history-195233
india pakistan	https://en.wikipedia.org/wiki/Victory_Day_of_Bangladesh https://en.wikipedia.org/wiki/Biharis_in_Bangladesh
	https://en.wikipedia.org/wiki/Gilgit-Baltistan

modified queries after applying Roccio algorithm to above queries:

Original query	**Query after roccio algorithm**
australia	australia map
china	china republic
ethiopia	ethiopia map
pakistan	pakistan kashmir
bangladesh area	bangladesh area code
pakistan terrorism	pakistan terrorism news
china population	china population policy
india pakistan	india Pakistan bbc
australai	australai coco
paksitan	paksitan east
chna	chna corona

germny	germny map
hindustan	hindustan time
britaan	britaan bhutan
franche	franche fries
gana	gana gana
indo	indo democrat
indo china	indo china wall
image argentina	imagining argentina wiki
iraq for sale	iraq sale people

The approach was used for the 20 queries mentioned above. The results obtained were not very promising and universal set of weights could not be arrived. Following were some of the findings:

1. Some queries fetched no documents – like "hindustan", "britaan" etc. which meant that the data we crawled did not contain neither the pages nor any reference to these words. In such cases, relevance model did not work
2. Some queries fetched all irrelevant documents – like "image argentina" fetched results for "imagining argentina movie". So the modified query was "image argentina satelite"
3. Some queries fetched great relevant results – like "china population" which fetched the china population 2020 and china child policy. In such cases, weights stabilized too early and so the modified query was not much relevant.
4. Performance issues as too much time was taken for some queries to get the stabilized weights, hence could not be performed on the fly
5. Each query needed its own unique set of weights

Due to the above issues, it was decided not to include the code for Rocchio algorithm for expanding the query, rather the results of the observations were used in the pseudo relevance feedback approaches to filter out the stop words, avoid spelling mistakes etc.

Pseudo Relevance Feedback:

1. **Association cluster**: The idea is that stems which co-occur frequently inside documents have a synonymity association
2. Metric cluster: The idea is that the stems which occur far apart in the document are less co-related to the terms that occur closer like in the same sentence.
3. Scalar cluster: The idea is that two stems are more co-related if they have similar neighbourhood.

Following is the list of 54 queries used for pseudo relevance feedback. (Each query run 3 times with 3 different cluster methods as mentioned above):

Query	**Relevant results fetched after expansion(pages)**	**Most relevant expanded query**	**Cluster method used**
china	10	china map republic	association, scalar
china population	10	china population policy child	association,
australia	10	australia coco demographic	association, scalar
bangladesh	9	bangladesh people arm	metric, association
ethiopia	2	ethiopia concur confluence	metric, association, scalar

india pakistan	8	india pakistan kashmir bangladesh war	metric, association
france	5	france wins wine map	metric, association, scalar
germany	5	germany squad language human	metric, association, scalar
france area	8	france area code people	association, scalar
germany flag	10	germany flag color red	association
china coronavirus	9	china coronavirus spread world	metric, association, scalar
pakistan terrorist	4	pakistan terrorist news kashmir	metric, association, scalar
qatar	9	qatar airway oil currency	metric, association,
canada weather	1	canada weather daily toronto canadian	metric, association, scalar
india	9	india map democracy	association
hindustan	0	hindustan article time	metric, association, scalar
gana	0	gana gana	metric, association
franche	0	franche firewatch	association, scalar

Based upon the above results, it was decided that the association cluster method is the best for implementation of the pseudo relevance feedback.

Some challenges:

1. Scalar clustering took too long to run so we can't perform them well
2. Metric clustering also was taking time in some cases.

Query 1: australia
local document set(ids):

491d54c4-885c-11ea-bc55-0242ac130003
491d571c-885c-11ea-bc55-0242ac130003
491d57e4-885c-11ea-bc55-0242ac130003
491d58a2-885c-11ea-bc55-0242ac130003
491d5ab4-885c-11ea-bc55-0242ac130003
491d5b90-885c-11ea-bc55-0242ac130003
491d5c58-885c-11ea-bc55-0242ac130003
491d5d16-885c-11ea-bc55-0242ac130003
491d5f00-885c-11ea-bc55-0242ac130003
491d5fd2-885c-11ea-bc55-0242ac130003
491d6090-885c-11ea-bc55-0242ac130003
491d614e-885c-11ea-bc55-0242ac130003
491d620c-885c-11ea-bc55-0242ac130003
491d62ca-885c-11ea-bc55-0242ac130003
491d6392-885c-11ea-bc55-0242ac130003
491d65cc-885c-11ea-bc55-0242ac130003
491d66a8-885c-11ea-bc55-0242ac130003
491d6766-885c-11ea-bc55-0242ac130003
491d682e-885c-11ea-bc55-0242ac130003
491d68ec-885c-11ea-bc55-0242ac130003
491d69b4-885c-11ea-bc55-0242ac130003

491d6a72-885c-11ea-bc55-0242ac130003
491d6b30-885c-11ea-bc55-0242ac130003
491d6d6a-885c-11ea-bc55-0242ac130003
491d6e32-885c-11ea-bc55-0242ac130003
491d6ef0-885c-11ea-bc55-0242ac130003
491d6fae-885c-11ea-bc55-0242ac130003
491d7076-885c-11ea-bc55-0242ac130003
491d71ac-885c-11ea-bc55-0242ac130003
491d7288-885c-11ea-bc55-0242ac130003

india_pakistan_local_doc.txt

local vocabulary set:

local_vocabulary_set1.txt

local stem set:

local_stem_set1.txt

Query 2: india Pakistan
local document set(ids):
867c15ba-885e-11ea-bc55-0242ac130003
867c193e-885e-11ea-bc55-0242ac130003
867c1b0a-885e-11ea-bc55-0242ac130003
867c1c72-885e-11ea-bc55-0242ac130003
867c1db2-885e-11ea-bc55-0242ac130003
867c1eca-885e-11ea-bc55-0242ac130003
867c205a-885e-11ea-bc55-0242ac130003
867c219a-885e-11ea-bc55-0242ac130003
867c229e-885e-11ea-bc55-0242ac130003
867c23fc-885e-11ea-bc55-0242ac130003
867c253c-885e-11ea-bc55-0242ac130003
867c264a-885e-11ea-bc55-0242ac130003
867c27d0-885e-11ea-bc55-0242ac130003
867c2906-885e-11ea-bc55-0242ac130003
867c2a1e-885e-11ea-bc55-0242ac130003
867c2b7c-885e-11ea-bc55-0242ac130003
867c2cc6-885e-11ea-bc55-0242ac130003
867c2e60-885e-11ea-bc55-0242ac130003
867c2faa-885e-11ea-bc55-0242ac130003
867c30fe-885e-11ea-bc55-0242ac130003
867c3252-885e-11ea-bc55-0242ac130003
867c3392-885e-11ea-bc55-0242ac130003
867c3518-885e-11ea-bc55-0242ac130003
867c36e4-885e-11ea-bc55-0242ac130003

867c38b0-885e-11ea-bc55-0242ac130003
867c39fa-885e-11ea-bc55-0242ac130003
867c3b3a-885e-11ea-bc55-0242ac130003
867c3c98-885e-11ea-bc55-0242ac130003
867c4350-885e-11ea-bc55-0242ac130003
867cac96-885e-11ea-bc55-0242ac130003

india_pakistan_local_doc.txt

local vocabulary set:

local_vocabulary_set2.txt

local stem set:

local_stem_set2.txt

Query 3: china population
local documents (ids) :
e5200812-8861-11ea-bc55-0242ac130003
e5200a10-8861-11ea-bc55-0242ac130003
e5200b0a-8861-11ea-bc55-0242ac130003
e520115e-8861-11ea-bc55-0242ac130003
e52014e2-8861-11ea-bc55-0242ac130003
e520160e-8861-11ea-bc55-0242ac130003
e52017d0-8861-11ea-bc55-0242ac130003
e5202130-8861-11ea-bc55-0242ac130003
e520241e-8861-11ea-bc55-0242ac130003
e520250e-8861-11ea-bc55-0242ac130003
e52025e0-8861-11ea-bc55-0242ac130003
e520269e-8861-11ea-bc55-0242ac130003
e520275c-8861-11ea-bc55-0242ac130003
e520281a-8861-11ea-bc55-0242ac130003
e52029b4-8861-11ea-bc55-0242ac130003
e5202a90-8861-11ea-bc55-0242ac130003
e5202b62-8861-11ea-bc55-0242ac130003
e5202c20-8861-11ea-bc55-0242ac130003
e5202e78-8861-11ea-bc55-0242ac130003
e5202f54-8861-11ea-bc55-0242ac130003
e520301c-8861-11ea-bc55-0242ac130003
e52030e4-8861-11ea-bc55-0242ac130003
e52031ac-8861-11ea-bc55-0242ac130003
e520326a-8861-11ea-bc55-0242ac130003
e5203328-8861-11ea-bc55-0242ac130003
e520362a-8861-11ea-bc55-0242ac130003
e5203710-8861-11ea-bc55-0242ac130003

e52037ce-8861-11ea-bc55-0242ac130003
e5203896-8861-11ea-bc55-0242ac130003
e520395e-8861-11ea-bc55-0242ac130003

china_population_local_doc.txt

local vocabulary set:

local_vocabulary_set3.txt

local stem set:

local_stem_set3.txt

from the table above, we can see that the relevant pages obtained are more when association cluster method is used and the time consumed for association cluster is less. As a result, we can say that association cluster is best suitable when performance is considered. It also takes care of spelling mistakes. The number of relevant pages obtained from table shows the correlations for the queries. It also shows the expanded query for each original query.

Collaboration with UI and relevance model:

1) From the API (URL) call we receive original query and the cluster method type to be used.
2) Based on these parameters, the relevant model provides top 50 documents to the query expansion python code. These results are then used as local documents to form local vocabulary , stem sets.
3) Based on the query parameter the cluster method out of association, metric and scalar is called to return expanded query to relevance model.
4) The relevance model then hits solar in order to received results for expanded query.
5) These results are then passed to front end and displayed.

Query selection for demo:

For the demo, the query “china population” was used since it has two words and can be enhanced with other words. It is tested for association cluster method.

7. Discussion

After completing this project, we have learned the many different aspects required to make a simple and working search engine. We were able to create a search engine that could take crawled webpages and return somewhat relevant results using the different techniques that we learned in class. The project gave us lots of practical knowledge of creating a search engine and also showed just that we have only really scratched the surface of Information Retrieval. When comparing the results that we worked so hard to get with large search engines such as Google and Bing, we realized that the work we did is small compared to what it takes to create and maintain popular and up-to-date search engines like Google or Bing.

8. Conclusion

In conclusion, for this project, we created a search engine — using the knowledge we gained throughout this class and some open source programs — that focuses on information related to Countries. The five of us splits the 5 different tasks required to build the search engine and collaborated to combine our individual parts into a final product that can provide relatively accurate information about Countries. While there were many challenges along the way, we were able to overcome them and create a good search engine. This project provided us with an opportunity to apply the knowledge we gained in this class in a practical way and boosted our understanding of the theoretical concepts we learned.